

The 2026 TTRPG PDF Layout Vulnerability Checklist

Why Standard AI Breaks Your Foundry VTT Layouts & How Deterministic Pipelines Fix It

Prepared by GM-Copilot Studios | B2B Technical Whitepaper & Audit Guide

Executive Summary

Converting backlist TTRPG rulebooks into digital VTT modules (Foundry VTT, Roll20) is one of the most labor-intensive bottlenecks facing game publishers today. While generic AI wrappers promise instant PDF parsing, they fail on complex tabletop layouts because probabilistic language models guess rather than verify. This checklist identifies the four primary structural failure points in TTRPG layout design and outlines how to audit your backlist before digital fulfillment.

The 4 Critical Layout Failure Points

1. Multi-Column Flow Traps

Standard PDF parsers read text left-to-right across the page geometry, ignoring vertical gutter boundaries. This causes multi-column stat blocks and spell descriptions to merge horizontally, rendering action descriptions completely incoherent.

2. Implied Table & Matrix Boundaries

TTRPG tables frequently rely on whitespace or subtle zebra-stripping rather than explicit border lines. Generative vision models routinely drop column alignments, shifting spell save DCs into creature movement speeds or merging nested weapon properties.

3. Stat Block Invariant Drift

A single hallucinated number in an Armor Class or Hit Die calculation invalidates a creature's Challenge Rating (CR). Generative models lack mathematical verification kernels to enforce game balance invariants during extraction.

4. Custom Trait & Inline Condition Tags

Modern layout design heavily utilizes custom inline icons, bolded action costs (e.g., 1-Action, Bonus Action), and inline condition tags. Unstructured parsers treat these as random noise or strip them entirely during plain text conversion.

Diagnostic Comparison

Extraction Vector	Standard AI Wrapper (LLM / RAG)	Deterministic Engine (OpenCV + Lean 4)
Layout Vision	Probabilistic bounding box guessing	OpenCV spatial contour & gutter mapping
Math Verification	None (Generative guesswork)	Lean 4 formal logic gate validation
Table Parsing	Collapses nested cells into string slop	Explicit cell-coordinate matrix reconstruction
IP & Data Security	Data trained or stored in third-party logs	100% in-memory processing & strict FinOps purge

Publisher 5-Minute Layout Audit Checklist

Use this quick checklist to evaluate your PDF catalog before attempting digital VTT automation:

- ☐ **Gutter Check:** Do multi-column pages have less than 0.25-inch margin gutters?
- ☐ **Borderless Tables:** Are spell matrices or equipment pricing tables rendered without explicit bounding gridlines?
- ☐ **Embedded Formulas:** Do creature stat blocks rely on derived attributes (e.g., HP calculated from Constitution mod + Hit Dice)?
- ☐ **Layered Artwork:** Does background texture or page artwork bleed beneath stat block text boxes?
- ☐ **Custom Typography:** Does your document use non-standard vector fonts for action icons or headers?

If you checked two or more boxes, generic LLM ingestion will result in structural failure and hallucinated JSON payloads.

Automate Your Backlist Deterministically

At **GM-Copilot Studios**, we eliminate 95% of manual VTT data entry through zero-hallucination, spatial-aware ingestion pipeline architecture. Stop manually typing stat blocks into Foundry VTT and Roll20.

Request a \$99 5-Page Structural Audit: Upload a sample PDF to receive a complete layout vulnerability report and verified JSON exports.

Website: cmccombs01.github.io/GM-Copilot-Studios